WEEK 06 TECHNICAL REPORT

YCB-V Error Analysis II

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Group	Group 1 — Basic ML Knowledge
Main Quest	Structuring Machine Learning Projects
Side Quest	Error Analysis II

1. Objective

The Week 06 objective was to convert the measured Week 05 error analysis into an evidence-based improvement plan and run targeted experiments rather than making random model changes. The notebook preserved the same five-class YCB-V task, the same BOP'19 subset, the same scene-aware evaluation logic, and the original Week 04 E1 baseline. Three candidate changes were chosen because each corresponded to a concrete failure hypothesis from Week 05: crop-context contamination, geometric distortion caused by square resizing, and loss of spatial structure in the flattened MLP.

The decision rule was defined before testing: compare every candidate against a fixed control on three deterministic scene-held-out development splits, use mean macro-F1 as the primary metric, keep the original Week 04 test scenes untouched during selection, and evaluate only the selected change once on that final held-out test split. This turns error analysis into a controlled decision process rather than a broad hyperparameter search.

2. Main Quest Progress and Relevant Concepts

The Main Quest remains Structuring Machine Learning Projects. Week 06 applies the ML-strategy ideas most directly connected to controlled improvement: orthogonalization, single-number selection metrics, train/dev/test distribution design, and prioritizing fixes that are supported by measured error patterns.

- 01 — Carrying Out Error Analysis
- 02 — Cleaning Up Incorrectly Labeled Data
- 03 — Build Your First System Quickly, then Iterate
- 04 — Training and Testing on Different Distributions
- 05 — Bias and Variance with Mismatched Data Distributions
- 06 — Addressing Data Mismatch
- 07 — Transfer Learning
- 08 — Multi-task Learning
- 09 — What Is End-to-End Deep Learning?
- 10 — Whether to Use End-to-End Deep Learning

2.1 Concepts directly connected to this Side Quest

- **Orthogonalization:** each experiment changes one causal lever while optimizer, learning rate, weight decay, dropout, class set, data filters, and selection rule remain fixed.
- **Dev/test distribution design:** experiment selection uses repeated whole-scene holdouts drawn only from the original train+validation scenes; the original two test scenes remain untouched until the final evaluation.
- **Single-number decision metric:** mean cross-scene macro-F1 is the primary selector, preventing a high overall accuracy from hiding class-specific failure.
- **Error-driven iteration:** the Week 05 confusion patterns define the hypotheses. Model changes are not introduced unless an observed failure motivates them.

3. Baseline, Scope, and Reproducibility

All Week 06 comparisons start from the exact Week 04 E1_regularization evaluation verified in Week 05. In this report, "test" means the internally held-out scene split used by the internship notebooks; it is not an official YCB-V benchmark submission. The uploaded Week 06 notebook was executed and contains the reported experimental outputs.

Setting	Configuration preserved for Week 06
Source	Lightweight YCB-V BOP'19 subset
Classes	tomato_soup_can (4); sugar_box (3); master_chef_can (1); tuna_fish_can (6); bleach_cleanser (12)
Usable-instance filter	Bounding box present; width and height >= 24 px; visible fraction >= 0.35
Original input	Ground-truth crop with 8% padding; direct RGB resize to 64 x 64; normalization mean/std = 0.5 / 0.5
Original split	877 train / 375 validation / 450 test instances; 8 / 2 / 2 scenes with no scene overlap
Control model	Flattened RGB -> 256 ReLU -> dropout -> 128 ReLU -> dropout -> five outputs
Fixed training settings	Adam, learning rate 1e-3, dropout 0.30, weight decay 1e-4; max 15 epochs; early stopping patience 4
Week 04 / 05 baseline	Held-out accuracy 41.78% (188/450); macro-F1 0.4224; validation-test accuracy gap about 57.16 percentage points

4. Decision Process and Method

Week 05 showed 262 errors on 450 held-out samples. The largest directed failures were tuna_fish_can -> tomato_soup_can (69), sugar_box -> bleach_cleanser (46), tomato_soup_can -> bleach_cleanser (45), bleach_cleanser -> tomato_soup_can (32), and tomato_soup_can -> tuna_fish_can (31). These patterns, together with the large validation-test gap, were converted into three testable hypotheses.

ID	Observed evidence / hypothesis	Single change
B0_control	Reference condition using the exact Week 04 preprocessing and regularized MLP.	No change: 8% padded crop, direct 64 x 64 resize, MLP.
X1_no_context_padding	Nearby-object/background context may be inside the padded crop and may not generalize across scenes.	Set crop padding from 8% to 0%; keep direct square resize and MLP.
X2_preserve_aspect_ratio	Direct square resizing may distort shape and viewpoint cues, especially across new scenes.	Keep 8% padding but resize with aspect-preserving letterbox to 64 x 64; keep the MLP.
X3_compact_cnn	The flattened MLP may be too sensitive to exact pixel positions and may discard useful spatial locality.	Keep baseline preprocessing but replace only the MLP with a small three-block CNN.

4.1 Predefined selection rule

- Build three deterministic development folds by holding out whole scenes from the original Week 04 train+validation scene pool.
- Evaluate B0, X1, X2, and X3 on the same folds; use mean macro-F1 as the primary metric and mean accuracy as a secondary metric.
- Track recall for the four problematic classes so an overall score cannot hide a regression.
- Do not use the original two Week 04 test scenes to choose the experiment.
- After selection, train the chosen configuration with the original train/validation protocol and evaluate once on the untouched test split.

5. Implementation

Notebook stage	What was executed	Purpose
Part A	Re-established exact Week 05 evidence: full confusion matrix, class metrics, split gap, and directed transitions.	Create an auditable starting point.
Part B	Mapped each observed failure to one hypothesis and one isolated intervention.	Prevent random or multi-factor model changes.
Parts C-D	Reproduced the same BOP data pipeline and created three whole-scene development folds.	Measure generalization before touching the final test set.
Parts E-F	Implemented controlled crop, resize, MLP/CNN options and fixed training utilities.	Hold non-tested variables constant.
Parts G-H	Ran all four configurations on the same three folds, selected by macro-F1, then ran one final held-out test.	Generate comparable evidence and preserve test integrity.
Part I	Converted the selected result into a next-step recommendation.	Continue the error-analysis loop using measured outcomes.

6. Results

6.1 Week 05 evidence carried into Week 06

The baseline evidence is unchanged: 188 of 450 predictions were correct and 262 were errors. Class recall was 49.33% for tomato_soup_can, 38.67% for sugar_box, 90.67% for master_chef_can, 8.00% for tuna_fish_can, and 14.67% for bleach_cleanser. The largest single failure was tuna -> tomato, affecting 69 of 75 tuna instances.

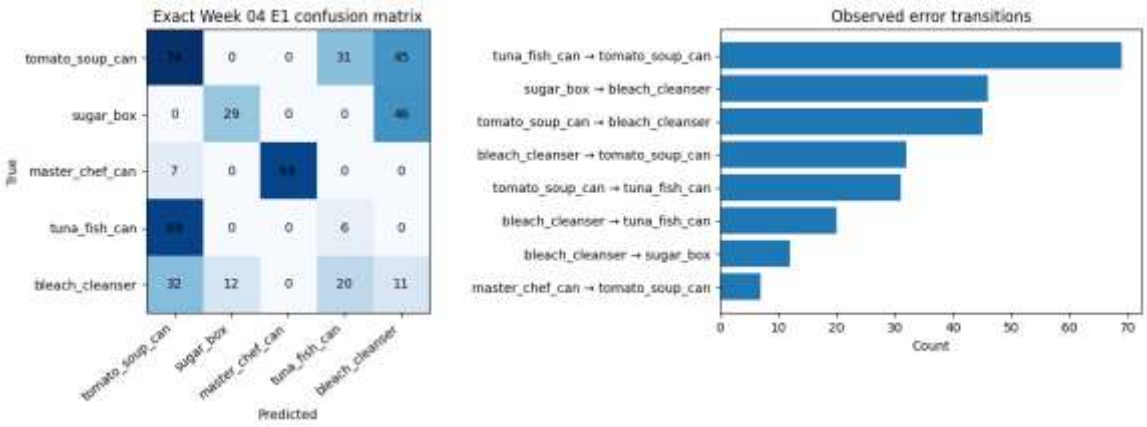


Figure 1. Exact Week 04 E1 confusion matrix and the error transitions that motivated the targeted Week 06 experiments.

6.2 Repeated scene-held-out development folds

Experiment	Fold 1 macro-F1	Fold 2 macro-F1	Fold 3 macro-F1	Mean	Std
B0_control	0.9867	0.8581	0.8151	0.8866	0.0893
X1_no_context_padding	0.9787	0.9960	0.8256	0.9334	0.0938
X2_preserve_aspect_ratio	0.9947	1.0000	0.9604	0.9850	0.0215
X3_compact_cnn	0.7223	0.8569	0.5707	0.7166	0.1432

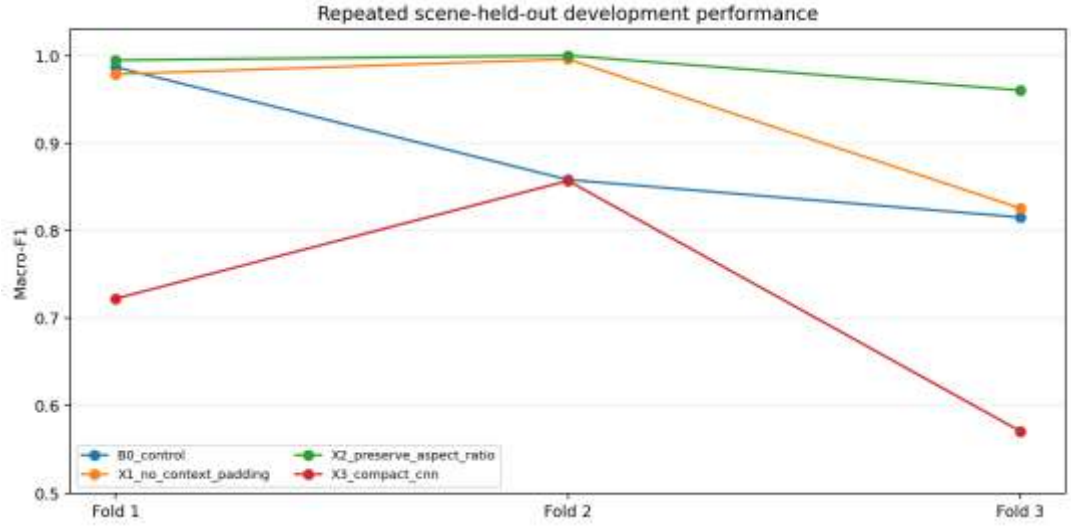


Figure 2. Macro-F1 for every targeted experiment on the same three scene-held-out development folds.

6.3 Aggregated experiment comparison

Experiment	Mean accuracy	Accuracy std	Mean macro-F1	Macro-F1 std	Delta F1 vs control
X2_preserve_aspect_ratio	0.9827	0.0255	0.9850	0.0215	+0.0984
X1_no_context_padding	0.9262	0.1058	0.9334	0.0938	+0.0468
B0_control	0.8689	0.1037	0.8866	0.0893	0.0000
X3_compact_cnn	0.7449	0.1006	0.7166	0.1432	-0.1699

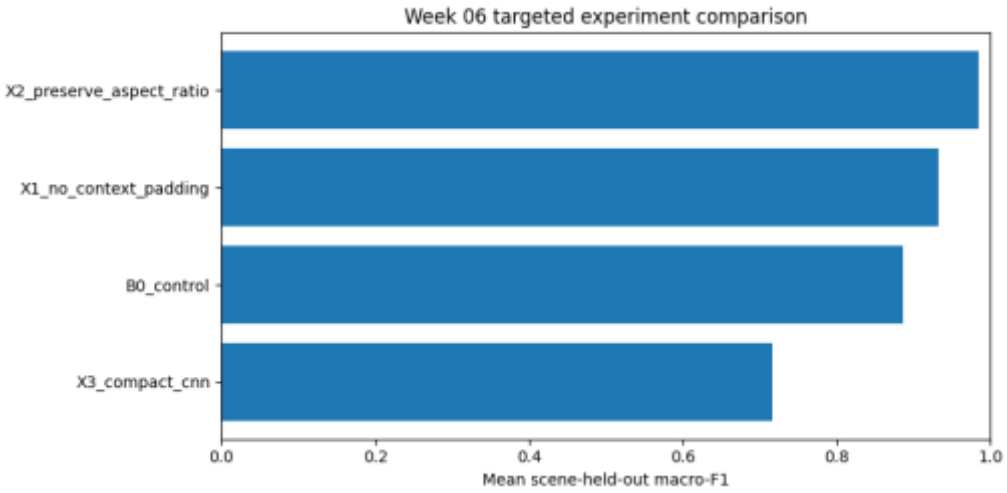


Figure 3. Notebook comparison of mean macro-F1 with variation across the three development folds. X2 is both the highest-scoring and the most stable candidate.

Selection: X2_preserve_aspect_ratio. It improved mean cross-scene macro-F1 by +0.0984 over the control and mean accuracy by +0.1138. Its macro-F1 standard deviation was 0.0215, substantially lower than B0 (0.0893), X1 (0.0938), and X3 (0.1432). The evidence therefore favors preserving object geometry during resizing rather than changing architecture.

X1 also improved the mean score, so crop context may still matter, but it was less stable and did not outperform X2. X3 performed worse than the original MLP under the fixed training protocol; simply switching to a CNN was not supported by these experiments.

6.4 Per-class development behavior

Experiment	Tomato recall	Sugar recall	Master recall	Tuna recall	Bleach recall
B0_control	0.9822	0.7844	0.8044	0.9778	0.9911
X1_no_context_padding	1.0000	0.9778	0.8267	0.9200	1.0000
X2_preserve_aspect_ratio	1.0000	1.0000	0.9533	0.9911	1.0000
X3_compact_cnn	0.9289	0.1667	0.8244	1.0000	0.9556

The selected X2 configuration did not win by sacrificing one of the problem classes: its mean development recall was 1.0000 for tomato, 1.0000 for sugar, 0.9533 for master, 0.9911 for tuna, and 1.0000 for bleach across the three folds.

6.5 Final held-out test of the selected experiment

Metric	Week 04 E1	Week 06 X2	Change
Correct / total	188 / 450	283 / 450	+95 correct predictions
Held-out accuracy	41.78%	62.89%	+21.11 percentage points
Macro-F1	0.4224	0.6468	+0.2244
Weighted F1	0.4263	0.6411	+0.2148
Errors	262	167	-95 errors

Class	Support	Week 04 recall	Week 06 recall	Delta	Week 06 precision	Week 06 F1
tomato_soup_can	150	0.4933	0.5000	+0.0067	0.7895	0.6122
sugar_box	75	0.3867	0.3867	0.0000	0.8788	0.5370
master_chef_can	75	0.9067	1.0000	+0.0933	0.7653	0.8671
tuna_fish_can	75	0.0800	1.0000	+0.9200	0.9036	0.9494
bleach_cleanser	75	0.1467	0.3867	+0.2400	0.2057	0.2685

The largest gain is tuna_fish_can: recall increased from 8.00% to 100.00%, eliminating the 69-case tuna -> tomato failure that dominated Week 05. Master recall also reached 100.00%, and bleach recall increased by 24 percentage points. Tomato recall changed little and sugar recall did not change on the final held-out scenes, so the selected preprocessing change does not solve every error family.

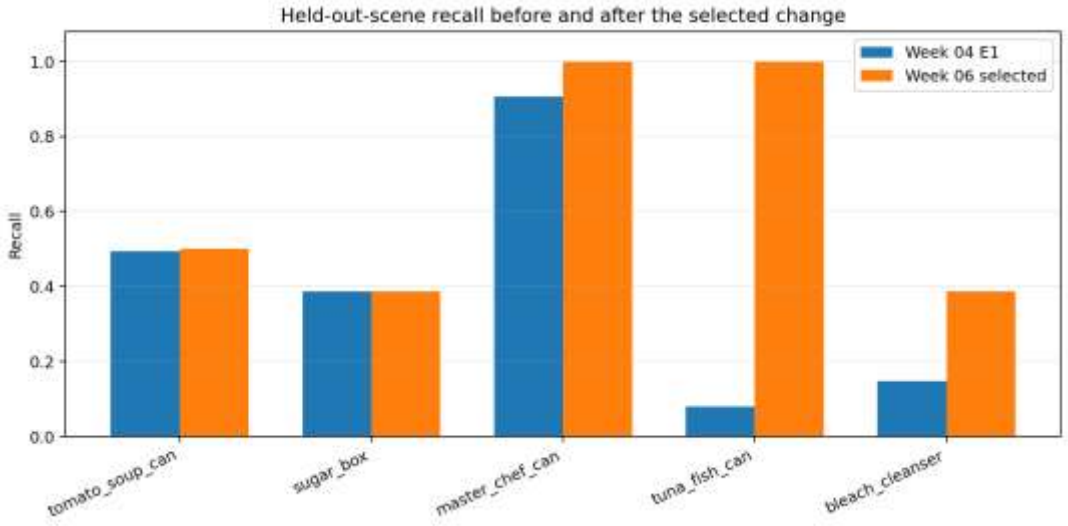


Figure 4. Held-out-scene recall before and after the selected aspect-preserving preprocessing.

6.6 Final confusion-matrix comparison

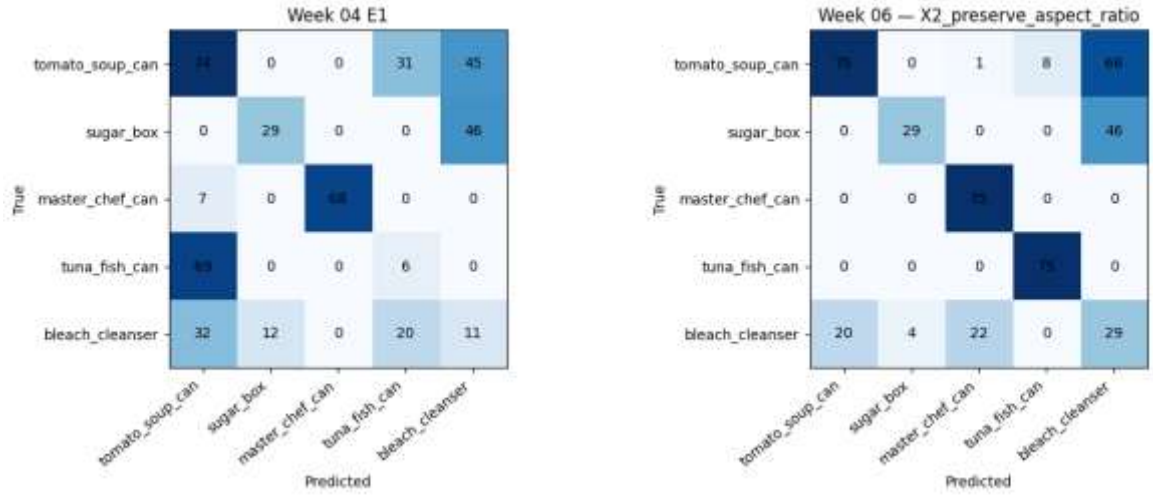


Figure 5. Original Week 04 E1 matrix versus the final Week 06 X2 matrix on the same 450 held-out examples.

True -> predicted	Week 04 count	Week 06 count	Interpretation
tuna -> tomato	69	0	Dominant Week 05 failure eliminated.
sugar -> bleach	46	46	No improvement on this failure family.
tomato -> bleach	45	66	Worsened; this becomes the largest remaining directed error.
bleach -> tomato	32	20	Improved by 12 cases.
tomato -> tuna	31	8	Improved by 23 cases.
bleach -> tuna	20	0	Eliminated.
bleach -> sugar	12	4	Improved by 8 cases.
master -> tomato	7	0	Eliminated.
bleach -> master	0	22	New concentration after the preprocessing change.

7. Failure Analysis and Interpretation

7.1 What the measurements establish

- **Geometry preservation is the best-supported lever tested this week.** X2 had the highest mean cross-scene macro-F1 (0.9850), the lowest macro-F1 variation (0.0215), and improved final held-out accuracy by 21.11 percentage points.
- **The Week 05 tuna/tomato failure was highly sensitive to preprocessing.** Tuna recall moved from 8.00% to 100.00%, and all 69 tuna -> tomato errors disappeared on the final held-out split.
- **The improvement is real but incomplete.** Final accuracy remains 62.89%, far below the near-perfect development results, so a scene/distribution mismatch or additional visual factors still remain.

- **Not every error family responds to the same intervention.** Sugar -> bleach stayed at 46 cases and tomato -> bleach increased from 45 to 66. Bleach also acquired a new 22-case confusion with master_chef_can.
- **Architecture change was not justified here.** Under the same training protocol, the compact CNN reduced mean macro-F1 by 0.1699 versus the control.

7.2 What the experiments do not prove

The selected result supports the hypothesis that direct square resizing was an important contributor to poor cross-scene transfer, but it does not prove that geometric distortion was the only cause. X1 also improved the development average, indicating that context may still influence generalization. The three deterministic development folds are useful for controlled comparison, but all three happened to include scene 55 in the held-out pair; more scene combinations would strengthen the estimate. The final test result also shows that unseen-scene difficulty remains substantial.

7.3 Evidence-based next-step recommendation

Keep X2_preserve_aspect_ratio as the new preprocessing baseline. Do not revert to direct square resizing and do not switch to the compact CNN based on the current evidence. The next experiment should change only one additional factor: add mild viewpoint/scale augmentation while preserving the aspect-ratio letterbox pipeline, then evaluate it with the same scene-held-out protocol.

The next error review should focus specifically on the two unresolved groups that now dominate the 167 remaining errors: tomato_soup_can -> bleach_cleanser (66) and sugar_box -> bleach_cleanser (46), plus the new bleach_cleanser -> master_chef_can pattern (22). These targeted groups provide a clearer basis for Week 07 than another generic optimizer or architecture sweep.

8. Conclusion and Next Step

Week 06 completed the required Error Analysis II loop: the Week 05 evidence was converted into explicit hypotheses, each hypothesis was mapped to a one-variable-at-a-time experiment, the candidates were compared on repeated scene-held-out development splits, and only the selected experiment was evaluated on the untouched final test scenes.

The evidence selected X2_preserve_aspect_ratio. Its mean cross-scene macro-F1 was 0.9850 compared with 0.8866 for the control, and its final held-out accuracy increased from 41.78% to 62.89%. The change removed the dominant 69-case tuna -> tomato failure and improved master and bleach recall, while leaving sugar -> bleach unresolved and worsening tomato -> bleach. This result demonstrates why targeted experiments are more informative than random model changes: one preprocessing change fixed a major error family, whereas the architecture change performed worse.

Recommended Week 07 direction: preserve the selected aspect-ratio-aware preprocessing and test mild viewpoint/scale augmentation as the next isolated intervention. Continue using whole-scene development folds, and prioritize visual review of the remaining tomato/bleach and sugar/bleach confusions before adding broader complexity.